TITLE OF THE PROJECT
SUMO ROBOT(ROBOTICS)

Abstract

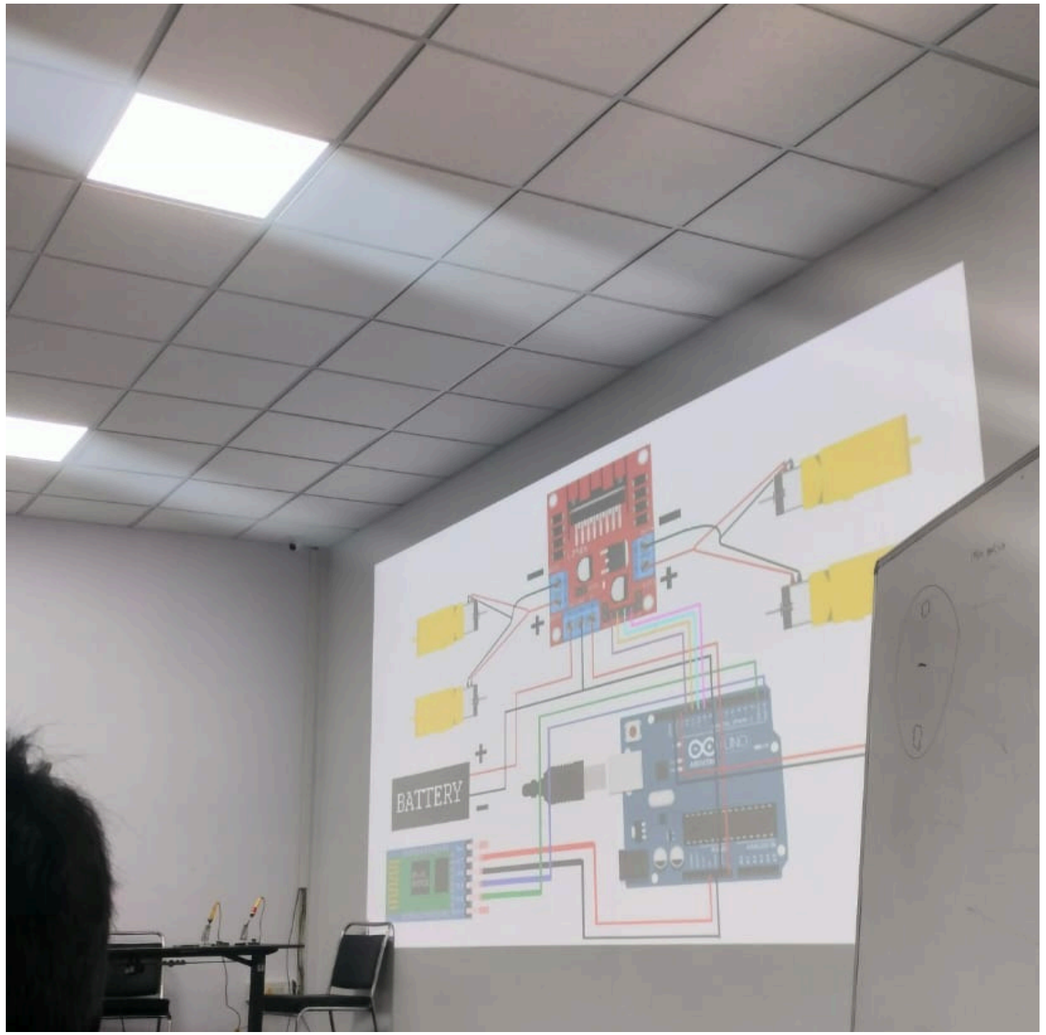
The Bluetooth Controlled Sumo Robot is a wireless robotic system designed to participate in sumo-style robot competitions by pushing the opponent out of the circular arena. The robot is controlled remotely through a smartphone or Bluetooth-enabled device using Bluetooth communication technology. The project combines embedded systems, motor control, wireless communication, and robotics to achieve efficient movement and real-time control.

Components Required

- Arduino Uno
- L298N Motor Driver Module
- HC-05 Bluetooth Module
- BO DC Gear Motors – 4 Motors
- Robot Chassis
- Wheels
- Battery
- Jumper Wires
- Power Connector / Switch
- Smartphone with Bluetooth Control App

Working principle

The Bluetooth Controlled Sumo Robot works by receiving wireless commands from a smartphone through the HC-05 Bluetooth module. The Arduino Uno processes these commands and controls the DC motors using the L298N motor driver module. Based on the input commands such as forward, backward, left, right, and stop, the robot moves accordingly inside the sumo arena. The battery supplies power to all components, enabling smooth movement and effective control during robot competitions.


BATTERY

Features

- Wireless control using Bluetooth
- Easy movement control through smartphone
- Fast response and smooth operation
- Compact and stable robot design
- High torque motor support
- Forward, backward, left, and right movement
- User-friendly operation
- Rechargeable battery powered
- Suitable for robotics competitions
- Low-cost and efficient system

Applications

- Robotics competitions
- Educational and training purposes
- Learning embedded systems and robotics
- Bluetooth communication applications
- Research and development projects
- Automation and control system studies
- Engineering mini projects
- Wireless robotic vehicle applications
- STEM and technical workshops
- Remote-controlled robotic systems

Conclusion

The Bluetooth Controlled Sumo Robot is an efficient and user-friendly robotic system designed for wireless control and robotic competitions. The project demonstrates the integration of Arduino, Bluetooth communication, motor control, and embedded systems to achieve smooth and accurate robot movement. It helps in understanding practical concepts of robotics, automation, and wireless technology while providing a low-cost and effective solution for educational and competition purposes.